

1 (2.11) verification

The admissible weight property

$$(G(t, \cdot) * \rho(\cdot))(x) = (2\pi t)^{-d/2} \int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) \rho(y) dy \leq K_T \rho(x) \quad \forall x \in \mathbb{R}^d, K_T \in \mathbb{N}, t \in [0, T]$$

tells us that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq K_T \quad t \in [0, T]. \quad (1.1) \quad \boxed{\text{aw}}$$

The expression for K_t should be

$$K_t := \sup_{x \in \mathbb{R}^d} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)}$$

The limit in (1.1) gives a sufficient (also necessary) condition for K_t to be finite, but in general, this limit is not equal to the value of K_t .

I meant for K_T to mean $C_\rho(T)$. I made some corrections above

We want to show that

$$\exp(-a|x|^2) \leq K'_a \rho(x) \quad (1.2) \quad \boxed{\text{con}}$$

Why does (1.1) imply (1.2)?

(1.4) and (1.1) imply (1.3) which implies (1.2)

and it suffices to show that

$$\lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq K'_a. \quad (1.3) \quad \boxed{\text{con1}}$$

since then

$$\exp(-a|x|^2) \leq (K'_a + \epsilon) \rho(x) \quad |x| > R_\epsilon$$

and by the positivity and continuity of both $\exp(-a|x|^2)$ and $\rho(x)$, we know that there exists some positive constant $K(a, \epsilon)$ such that

$$\exp(-a|x|^2) \leq K'(a, \epsilon) \rho(x) \quad x \leq R_\epsilon$$

and by then letting $K(a, \epsilon) = \max\{K'(a, \epsilon), (K'_a + \epsilon)\}$ then

$$\exp(-a|x|^2) \leq K(a, \epsilon) \rho(x) \quad x \in \mathbb{R}^d$$

which is what we want to show.

In order to show (1.3), it suffices to show that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\exp(-a|x|^2)} = \infty \quad (1.4) \quad \boxed{\text{con2}}$$

because this would mean that $\exp(-a|x|^2)$ is much smaller than $(G(t, \cdot) * \rho(\cdot))(x)$ for all large $|x|$ and since (1.1) is true then this would mean that (1.3) would be true as well by squeeze theorem,

$$0 \leq \lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq \lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq K_t$$

(1.4) is proved as follows: first recall that $|\rho(x)| < M$ for some $M \in \mathbb{N}$. Then,

$$0 \leq \frac{\int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) \rho(y) dy}{\exp(-a|x|^2)} \leq \frac{M \int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) dy}{\exp(-a|x|^2)}$$

$$\begin{aligned}
&= \frac{M * \sqrt{2\pi t}}{\exp(-a|x|^2)} \\
&\rightarrow \infty \quad \text{as } |x| \rightarrow \infty
\end{aligned}$$

Thus (1.4) is proved which means that (1.2) is also true.

Can you give an explicit form of $C_\rho(T)$ so that we can check whether $\sup_{T>0} C_\rho(T) < \infty$, or your K_t ? If not, can you derive some sufficient conditions that guarantee this property?

In the above wrote K_t to mean the same thing as $C_\rho(T)$. However in general there is no explicit form for this constant and also in general, $\sup_{T \geq 0} K_T \not< \infty$. For example, when we consider the weight $\rho(x) = \exp(-|x|)$ in $\mathbb{R}$ then we can show that $(\bar{G}(t, \cdot) * \exp(-|\cdot|))(x) \leq 2 \exp(t/2) \exp(-|x|)$ and clearly $\sup_{t \geq 0} 2 \exp(t/2) = \infty$.

This is a good example. We should not expect $\sup_{T>0} C_\rho(T) < \infty$.

2 Mistake involving the $G(t, x) \leq KG(1, x)$ inequality

During a meeting we had, we proved that $G(t, x) \leq KG(1, x)$ where the constant K does not depend on x or t but we made a mistake in doing so. What we did is the following (for now we work in $\mathbb{R}$):

$$\begin{aligned}
\frac{G(t+1, x)}{G(1, x)} &= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2(t+1)} + \frac{|x|^2}{2}\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \left(-\frac{1}{t+1} + 1\right)\right) \quad \text{mistake was made here when factoring} \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \frac{t}{t+1}\right) \\
&\leq K
\end{aligned}$$

However if we fix the mistake, what we should get is that

$$\begin{aligned}
\frac{G(t+1, x)}{G(1, x)} &= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2(t+1)} + \frac{|x|^2}{2}\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \left(\frac{1}{t+1} - 1\right)\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(\frac{|x|^2}{2} \frac{t}{t+1}\right) \\
&\not\leq K
\end{aligned}$$

Let's remove the property $\sup_{t \geq 0} \mathbb{E} \left(|u(t, \cdot)|_\rho^2 \right) < \infty$ unless you can show that $C_\rho(T)$ is bounded in T .

3 Mistake that $G(t + t_0, x)$ is bounded above by $G(t, x)$ for $t > t_0$

$$\begin{aligned}
\frac{G(t + t_0, x)}{G(t_0, x)} &= \sqrt{\frac{t}{t + t_0}} \exp\left(-\frac{|x|^2}{2(t + t_0)}\right) \exp\left(\frac{|x|^2}{2t_0}\right) \\
&= \sqrt{\frac{t_0}{t + t_0}} \exp\left(\frac{|x|^2}{2(t + t_0)}\right)
\end{aligned}$$

and this is not bounded above by a constant independent of x and t .

This makes more sense!